Professional Self-Assessment

Earl J. Pfau

Department of Computer Science, Southern New Hampshire University

CS-499: Computer Science Capstone

Professor Joseph Conlan

October 17, 2026

- How my coursework has prepared and shaped me in my professional readiness.

While completing the computer science program and building my ePortfolio has given me a clear understanding of my strengths, my professional values, and the direction I want to take in the field of computer science. While I finished each course throughout the program it has contributed to developing practical, industry-aligned skills while also shaping how I approach problems, collaborating with others, and communicating technical ideas to others. I am going to discuss these five areas of computer science: collaborating in a team environment, communicating with stakeholders, data structures and algorithms, software engineering and database, and security.

One of the most impactful areas for growth for me was collaborating in a team environment. Through multiple courses I learned how to divide responsibilities, manage version control, and maintain consistent communication. Working with peers has taught me how to balance technical contributions with interpersonal skills: resolving conflicts, clarifying requirements, and ensuring that everyone's work integrated smoothly in the current projects. These were some of the experiences that helped me understand the importance of reliability, following industry standards, transparency, and adaptability in professional engineering teams.

I have also strengthened my ability to communicate with stakeholders, both technically and non-technically. Several of the courses require presenting project proposals, writing technical documentation and explaining design decisions to instructors. Not only that, a few of the courses we had to gather requirements through interviews with the stakeholders through watching video recordings and transcripts. Any courses that did not require stakeholder interactions were given a requirements document that outlined what the program must do. These experiences taught me how to translate complex concepts such as algorithmic trade-offs or even database schema decisions into clear, accessible explanations. This skill is essential in real-world environments where engineers must justify choices, gather requirements, and ensure alignment with business goals.

Another major part of my development came from consistently working with data structures and algorithms. The courses taught me how to think critically about performance, scalability, and efficiency. I have learned how to evaluate time and space complexity by choosing appropriate data structures for specific problems, all while keeping the design of my algorithms in balance with correctness and optimization. I have been able to apply these concepts outside the classroom as well, I have applied these principals to personal projects, such as building gaming servers, modifying open-source mods for the server, optimizing server settings and mods for performance, and experimenting with

different algorithmic approaches to solve practical problems. This reinforced my confidence in tackling complex problems and challenges.

Every bit of my coursework in software engineering and databases has honed my understanding in how to build complete, maintainable systems. I have gained experience with full development lifecycles: gathering requirements, designing architectures, implementing features, testing functionality, and documenting the results. The database courses have taught me how to design relational schemas, write efficient queries, and understand normalization, indexing, and data integrity. These skills have directly supported my ability to work on real applications that require structured data, backend logic, and long-term maintainability.

Finally, the program has strengthened my security mindset as well. Through security focused coursework, I have learned how to identify vulnerabilities, anticipate adversarial behavior, and design systems that protect data and resources. I have gained a great deal of experience with secure coding practices, authentication and authorization concepts, and threat modeling. This mindset has influenced how I approach every project, I now consider not only how a system works, but how it might fail or be exploited in the real world.

Overall, completing the program and developing my ePortfolio has helped me recognize my strengths in problem-solving, communication, teamwork, and technical execution. It also clarified my professional goals to build reliable, secure, and efficient software systems while continuing to grow as a collaborative engineer. The ePortfolio serves as a curated representation of these abilities and demonstrates my readiness to enter the computer science field as a capable and employable professional.

- How my artifacts fit together to represent my full range of skill.

The artifacts in my ePortfolio were selected to represent the breadth of my abilities across the core areas of computer science, and within my personal life. Each artifact highlights a different aspect and area of my technical skillset, and together they form a cohesive picture of my growth throughout my academic journey and within my personal life.

The following ePortfolio includes work related to software engineering, demonstrating my ability to design and implement structured, maintainable applications that follow industry standards. It also shows my ability in data structures and algorithms, where I applied theoretical concepts to solve real world practical problems efficiently. My database artifact highlights my understanding of how to incorporate real world projects into projects that typically don’t support certain kinds of databases, for example using Json as a database in

a C++ project. This highlights my understanding of database design, reading and writing to a database all while ensuring the integrity of the database. More importantly the artifact shows my ability to identify vulnerabilities and apply secure coding principles.

By showcasing these artifacts together, it strengthens my ePortfolio in illustrating how my skills have developed over the computer science program and how my skills complement one another. It shows that I can not only write code, but also design systems, optimize performance, secure code, and communicate my decisions effectively in a professional manner. These artifacts collectively demonstrate my readiness to contribute to real-world engineering teams and my ability to apply computer science principles to meaningful, industry aligned solutions.